

Wang Proposition 5.2

Mu-Tao Wang, <i>Mean Curvature Flow of Surfaces in Einstein Four-Manifolds</i> , Section 5,			
Proposition 5.2	323--324	Gaussian density	1
		varifold	“
” “multiplicity one”		Gaussian density	2
density	1	White	

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Danus Wang Huisken White

Type-I $\implies$ $\implies$ $\implies \Theta(y_0, t_0) = 1 \implies$.

Type-I	$ A^\lambda $		
		Huisken	
	multiplicity one		sheets
	Gaussian density 1	m	density m
density 1		White	

Proposition 5.2

multiplicity one Kähler Huisken

1.

Wang Proposition 5.2

$$F:\Sigma\times[0,t_0)\longrightarrow M^4\hookrightarrow\mathbb{R}^N,$$

$$\eta_t= *F_t^*\omega>\delta>0,\qquad |A_t|^2\leq \frac{C}{t_0-t}.$$

(1.1)

F

t_0

immersions

Proposition 5.2

embeddedness

Theorem D

embedded

“

”

$\lambda_i\rightarrow\infty$

$s_i\rightarrow-1$

$\Sigma_{s_i}^{\lambda_i}$

Huisken [3]

$\Sigma_{s_i}^{\lambda_i}\longrightarrow\Sigma_{-1}^\infty$

(1.2)

$F^\perp=0$

multiplicity-one plane

Gaussian integral

1

(1.1)--(1.2)

2.

(y_0,t_0)

$F^\lambda(x,s)=\lambda\big(F(x,t_0+\lambda^{-2}s)-y_0\big),\qquad s<0.$

(2.1)

Σ_s^λ

λ

λ^{-1}

$|A^\lambda|^2(x,s)=\lambda^{-2}|A|^2\left(x,t_0+\frac{s}{\lambda^2}\right).$

(2.2)

(1.1)

$t_0+s/\lambda^2\in[0,t_0)$

$|A^\lambda|^2(x,s)\leq\lambda^{-2}\frac{C}{t_0-(t_0+s/\lambda^2)}$

$=\frac{C}{-s}.$

(2.3)

$$[a,b]\Subset (-\infty,0),$$

$$\lambda$$

$$\sup_{s\in[a,b]}|A^\lambda|^2\leq \frac{C}{-b}.$$
(2.4)

$$t\uparrow t_0$$

$$|A|^2(x,t)=\lambda^2|A^\lambda|^2(x,s)$$

$$(t_0-t)^{-1}$$

“

”

3. “ ”

$$(2.4)$$

$$|\nabla^k A^{\lambda_i}|\leq C_{k,K}$$
(3.1)

$$C^\infty$$

Huisken 1990 Proposition 2.3

Proposition 3.4

Brakke

1.

2.

$$m$$

3. **degree-one**

sheet

integral varifold

$$m$$

multiplicity one

sheet

4.

Wang

$$s_i\rightarrow -1$$

$$Q:=(h_{31k}-h_{42k})^2+(h_{32k}+h_{41k})^2=0$$
(4.1)

$$(4.1)$$

$$H=0,\qquad \nabla\eta=0.$$
(4.2)

Huisken

shrinker

$$H+\frac{1}{2}F^\perp=0. \tag{4.3}$$

(4.2)--(4.3)

$$F^\perp=0. \tag{4.4}$$

(4.4)

$$P$$

$$\operatorname{spt} V_{-1}=P \tag{4.5}$$

integral varifold

$$V_{-1}=m|P|,\qquad m\in\mathbb{N}, \tag{4.6}$$

(4.1)--(4.5) \hspace{1cm} m

$$m\geq 2\qquad \text{varifold }m|P|$$

$$A=H=F^\perp=\nabla\eta=Q=0.$$

5.

5.1

$$P=\{z=0\}\subset\mathbb{R}^3$$

$$\varepsilon_i\downarrow 0,\qquad R_i\uparrow\infty.$$

$$\Gamma_i$$

$$r\leq R_i$$

$$z=\varepsilon_i,\qquad z=-\varepsilon_i. \tag{5.1}$$

$$R_i\leq r\leq R_i+L$$

$$z=1\qquad z=-1$$

$$R_i+L$$

$$k\qquad i\qquad C_k$$

$$\sup_{\Gamma_i}|\nabla^kA_{\Gamma_i}|\leq C_k. \tag{5.2}$$

$$\Gamma_i$$

$$S^2$$

$$K\subset\mathbb{R}^3\qquad i\qquad K$$

(5.1)

$$\Phi\qquad \text{varifold}$$

$$i$$

$$\Phi$$

$$\begin{aligned} &\int_{\Gamma_i} \Phi(x,T_x\Gamma_i)\,d\mathcal{H}^2(x) \\ &= \int_P \Phi(y+\varepsilon_i\nu,P)\,d\mathcal{H}^2(y) + \int_P \Phi(y-\varepsilon_i\nu,P)\,d\mathcal{H}^2(y), \end{aligned}\tag{5.3}$$

$$\begin{array}{c} \Phi \\ \\ |\Gamma_i|\longrightarrow 2|P| \quad \ll \text{varifold} \gg. \end{array}\tag{5.4}$$

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-
-
-

sheets
 $\quad i \quad \quad i \quad$
 sheet separation

5.2

Γ_i

Wang	multiplicity one
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Proposition 5.2
 $\quad$
 symplectic

$P \subset \mathbb{C}^2$
 $\quad \quad \quad \eta = 1 \quad m|P|$

Kähler
 $\quad \quad \quad m$

6. Gaussian density

$$s=-1$$

$$G(x)=\frac{1}{4\pi}e^{-|x|^2/4}.\tag{6.1}$$

P

$$\begin{aligned} \int_P G\,d\mathcal{H}^2 &= \frac{1}{4\pi} \int_0^{2\pi} \int_0^\infty e^{-r^2/4} r\,dr\,d\theta \\ &= 1. \end{aligned}\tag{6.2}$$

$$V_{-1}=m|P|$$

$$\Theta(y_0,t_0)=\int G\,d\|V_{-1}\|=m\int_P G\,d\mathcal{H}^2=m. \tag{6.3}$$

$$\mathbb{I} \mathbb{I} \mathbb{I} \mathbb{I} + \mathbb{I} \mathbb{I} \, m \Longrightarrow \Theta = m, \tag{6.4}$$

$$“\hspace{1.5cm}\Theta=1”$$

$$m\geq 2\hspace{1cm}\begin{matrix}\text{White}&&\text{White}&&\text{Gaussian density ratio}&&1\\(6.3)\end{matrix}$$

7. density 2

7.1

$$F:\Sigma\times[0,t_0)\longrightarrow M^4$$

$$\text{K\"ahler}$$

$$*F_t^*\omega\geq\delta>0,\qquad |A_t|^2\leq\frac{C}{t_0-t}.\tag{7.1}$$

$$\text{Wang}\hspace{15cm}\text{tangent flow}$$

$$\text{Gaussian density}$$

$$\Theta(y_0,t_0)<2.\tag{7.2}$$

$$\varepsilon>0$$

$$[0,t_0+\varepsilon).$$

$$\begin{matrix} & (y_0,t_0) & (7.1) & & \text{tangent flow} \\ -1 & P & \text{varifold} \end{matrix}$$

$$V_{-1}=m|P|,\qquad m\in\mathbb{N}.\tag{7.3}$$

$$\text{Gaussian}\hspace{10cm}(6.3)$$

$$\Theta(y_0,t_0)=m.\tag{7.4}$$

$$(7.2)\hspace{1cm}m\hspace{1cm}m<2$$

$$m=1,\qquad \Theta(y_0,t_0)=1.\tag{7.5}$$

$$\varepsilon_{\mathrm{W}}>0\hspace{1cm}\text{White}$$

$$1 < 1 + \varepsilon_W,$$

$$\text{White} \qquad (y_0,t_0)$$

$$\Sigma \qquad \qquad \qquad t_0$$

$$\sup_{\Sigma\times [t_0-\tau,t_0)}|A|<\infty \tag{7.6}$$

$$\tau>0$$

$$\sup_{\Sigma\times [0,t_0)}|A|<\infty.$$

$$\varepsilon>0\qquad\qquad\qquad [0,t_0+\varepsilon)$$

$$7.2$$

$$(7.2) \qquad m\geq 2$$

1. tangent flow

unit-weight plane
2.

degree-one
3. varifold

$|P|$
4.

Kähler

$$\text{Bamler--Kleiner} \qquad \mathbb{R}^3$$

$$8. \quad \text{“} \qquad \qquad \text{”}$$

$$K\Subset P$$

$$X_i(p)=p+u_i(p),\qquad \|u_i\|_{C^k(K)}\longrightarrow 0\quad \text{as}\quad k.\tag{8.1}$$

$$\text{Jacobian } J_i \qquad 1 \qquad \qquad \phi$$

$$\int_{\Sigma_i}\phi\,d\mathcal{H}^2=\int_P\phi(X_i(p))J_i(p)\,d\mathcal{H}^2(p)\longrightarrow\int_P\phi(p)\,d\mathcal{H}^2(p).\tag{8.2}$$

$$\text{Gaussian}$$

$$\int_{\Sigma_i}G\,d\mathcal{H}^2\longrightarrow\int_PG\,d\mathcal{H}^2=1.\tag{8.3}$$

$$\text{“} \qquad \qquad \text{density } 1\text{”} \qquad \qquad \textbf{degree-one} \qquad \qquad \text{Gaussian}$$

$$\text{“} \qquad \qquad \qquad \qquad \qquad \qquad \qquad \text{”}$$

9.

(y_0, t_0)

$D_\lambda(y, t) = (\lambda(y - y_0), \lambda^2(t - t_0))$

(9.1)

- (y_0, t_0) multiplicity-one density 1
- density 0
- shrinker

Wang y_0 density
blow-up White

10. weighted Gaussian density

7

Gaussian density

$$\Theta^w = \int w \, G \, d\|V\|,$$

- $w \geq 1$ $\cos^{-p} \alpha$ $m \geq 2$ $\Theta^w \geq 2$ $\Theta^w < 2$
tangent density
- $w = 1 - \eta$ $w = 0$ m “weighted density 2”
 $m \geq 2$
weighted density Wang Gaussian
density

11.

1. “Type-I ”
2. “ ”
3. “ ”
4. “ ”
5. “ Gaussian integral 1 White ”

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